CSE 320 - Computer Networks B

LAB Session 2

02.04.2026

Packet Tracer:
https://www.packettracernetwork.com/download/download-packet-tracer.html
https://skillsforall.com/resources/lab-downloads?courseLang=en-US

Packet Tracer Everywhere: https://github.com/PTAnywhere/ptAnywhere-installation

1. Learn about the Link Layer's operations, focusing on Ethernet switching, VLAN configuration, and inter-VLAN routing in a simulated network environment using Cisco Packet Tracer.

a. Step 1: Open Cisco Packet Tracer

Launch Cisco Packet Tracer on your computer. If you don't have it installed, you need to download and install it from Cisco Networking Academy's website.

b. Step 2: Place Devices

Add Switches:

- Go to the bottom left corner where the device types are listed.
- Select Switch and then choose a generic switch (e.g., 2960).
- Click on the workspace to place two switches.

Add PCs:

- Select End Devices from the device types.
- Choose a generic PC and click on the workspace to place four PCs.

c. Step 3: Create the Network

Wire the Devices:

- Click on the Connections type (represented by a lightning bolt or cable icon).
- Select an appropriate cable (Auto Connect is recommended for simplicity).
- Click on a PC, then click on a port (FastEthernet is common), and connect it to one of the switches. Repeat this step for each PC, distributing them between the two switches as you see fit.

d. Step 4: Configure PC IP Addresses

- Click on a PC to open its configuration window.
- Go to the **Desktop tab**, then click IP Configuration.
- Enter an IP address and subnet mask. For simplicity, use the following scheme:
  - PC0: IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0
  - PC1: IP Address: 192.168.1.3, Subnet Mask: 255.255.255.0
  - PC2: IP Address: 192.168.1.4, Subnet Mask: 255.255.255.0
  - PC3: IP Address: 192.168.1.5, Subnet Mask: 255.255.255.0
- Make sure each device is in the same subnet (192.168.1.0/24 in this example) to ensure they can communicate.

e. Step 5: Testing Connectivity

- After configuring the IP addresses, test the network connectivity using the Ping command.
- Open the command prompt on a PC by clicking it, going to the Desktop tab, and selecting Command Prompt.
- Type ping [destination IP], for example, ping 192.168.1.3 from PC0 to test connectivity to PC1.
- If everything is configured correctly, you should see replies returning, indicating successful communication between the devices.

f. Step 6: Observing the Packet Flow (Optional)

- Choose a simple packet, click on a source PC, then click on a destination PC to generate a packet.Step 7: Save Your Work

Don’t forget to save your Packet Tracer file by going to File > Save As, and give it a meaningful name.

2. Another Simple Example:

This will require configuring multiple VLANs on the switches, setting up a router to handle inter-VLAN routing, and ensuring PCs on different VLANs can communicate. This scenario is excellent for understanding VLANs, inter-VLAN routing, and basic router configurations.

Create a network with two VLANs, configure a router for inter-VLAN routing, and ensure PCs on different VLANs can communicate.

Equipment Used:

- 2 Switches (Cisco 2960)
- 1 Router (Cisco 2911)
- 4 PCs

a. Step 1: Design the Network

- Open Cisco Packet Tracer and start a new project.
- Place two switches and one router on the workspace.
- Add four PCs, connecting two to each switch.
- Switch1 connected to Router's GigabitEthernet0/0 interface.
- Switch2 connected to Router's GigabitEthernet0/1 interface.
- PCs connected to switches, distributed across VLANs 10 and 20.

b. Topology assumptions

Left side (Switch0):

- PC0 → Switch0 Fa0/2 (VLAN 10)
- PC1 → Switch0 Fa0/3 (VLAN 20)
- Switch0 Fa0/1 → Router0 Gi0/0 (TRUNK)

Right side (Switch1):

- PC2 → Switch1 Fa0/2 (VLAN 10)
- PC3 → Switch1 Fa0/3 (VLAN 20)
- Switch1 Fa0/1 → Router0 Gi0/1 (TRUNK)

(If your PC ports differ, just apply the access-port config to the correct interfaces.

c. Addressing plan

Site A (Left / Router Gi0/0)

- **VLAN 10:** 172.16.10.0/24 **Gateway:** 172.16.10.1
- **VLAN 20:** 172.16.20.0/24 **Gateway:** 172.16.20.1

Site B (Right / Router Gi0/1)

- **VLAN 10:** 192.168.10.0/24 **Gateway:** 192.168.10.1
- **VLAN 20:** 192.168.20.0/24 **Gateway:** 192.168.20.1

PCs

- **PC0 (Left VLAN10):** 172.16.10.10 /24 **Gateway:** 172.16.10.1
- **PC1 (Left VLAN20):** 172.16.20.10 /24 **Gateway:** 172.16.20.1
- **PC2 (Right VLAN10):** 192.168.10.10 /24 **Gateway:** 192.168.10.1
- **PC3 (Right VLAN20):** 192.168.20.10 /24 **Gateway:** 192.168.20.1

d. Switch0 configuration

Create VLANs + assign access ports + trunk uplink

```
enable
configure terminal
hostname SW-LEFT

vlan 10
 name STAFF
vlan 20
 name STUDENTS
exit

interface fa0/2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
exit

interface fa0/3
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
exit

interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20
exit

end
write
```

Quick Checks

```
show vlan brief
show interfaces trunk
```

e. Switch1 configuration (Right)

```
enable
conf t
hostname SW-RIGHT

vlan 10
 name STAFF
vlan 20
 name STUDENTS
exit

interface fa0/2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
exit

interface fa0/3
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
exit

interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20
exit

end
wr
```

Quick Checks

```
show vlan brief
show interfaces trunk
```

f. Router0 configuration (2911) — router-on-a-stick on BOTH links

On a 911 in Packet Tracer, you typically use Gi0/0 and Gi0/1 with 802.1Q subinterfaces.

```
enable
conf t
hostname R-EDGE

! ---- Left trunk: Gi0/0 (Campus-A) ----
interface gi0/0
 no shutdown
exit
interface gi0/0.10
 encapsulation dot1Q 10
 ip address 172.16.10.1 255.255.255.0
exit
interface gi0/0.20
 encapsulation dot1Q 20
 ip address 172.16.20.1 255.255.255.0
exit

! ---- Right trunk: Gi0/1 (Campus-B) ----
interface gi0/1
 no shutdown
exit
interface gi0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gi0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
end
wr
```

Quick Checks

```
show ip interface brief
show running-config | section interface
show ip route
```

g. PC configuration (Packet Tracer GUI)

- **PC0: IP 172.16.10.10 / Mask 255.255.255.0 / GW 172.16.10.1**
- **PC1: IP 172.16.20.10 / Mask 255.255.255.0 / GW 172.16.20.1**
- **PC2: IP 192.168.10.10 / Mask 255.255.255.0 / GW 192.168.10.1**
- **PC3: IP 192.168.20.10 / Mask 255.255.255.0 / GW 192.168.20.1**

If any of these fail, your issue is almost always:

- wrong access VLAN on the switch port,
- trunk not trunking,
- router subinterface down / missing encapsulation,
- wrong default gateway on the PC.

Verification (unchanged, just new IPs)

1. Ping gateways:
   - PC0 → `ping 172.16.10.1`
   - PC1 → `ping 172.16.20.1`
   - PC2 → `ping 192.168.10.1`
   - PC3 → `ping 192.168.20.1`

2. Inter-VLAN left:
   - PC0 → `ping 172.16.20.10`
3. Cross-campus routing:
   - PC0 → `ping 192.168.10.10`
   - PC1 → `ping 192.168.20.10`

Take Home:

***Objective*:** Create a network with two VLANs, enable DHCP on a router to serve IP addresses to each VLAN, implement inter-VLAN routing, and secure the network using ACLs to restrict access between VLANs.

Equipment Needed:
2 Switches (e.g., Cisco 2960)
1 Router (e.g., Cisco 2911)
Multiple PCs (at least 2 per VLAN)

Connections (Ethernet cables)

Tasks:

Task 1: Create and Assign VLANs

Configure two VLANs on each switch. For example, VLAN 10 (Finance) and VLAN 20 (HR).

Assign switch ports to each VLAN. Ensure that each PC is connected to its respective VLAN.

Task 2: Configure the Router for DHCP and Inter-VLAN Routing

Set up the router's sub-interfaces for each VLAN, ensuring the correct encapsulation configuration.

Enable DHCP on the router: Define DHCP pools for each VLAN.

Specify the network, default router, and DNS server for each pool.

Task 3: Implement Inter-VLAN Routing

Ensure the router is configured to route traffic between VLANs using the router-on-a-stick configuration you set up with sub-interfaces.

Task 4: Secure the Network with Access Control Lists (ACLs) (Optional)

Create an ACL that permits or denies specific traffic. For example, allow Finance VLAN to access resources in the HR VLAN but deny the opposite.

Apply the ACL to the appropriate router interface in the inbound or outbound direction.

Show your work with a screen recorder and upload the video to YouTube/Google Drive (it can be private, but make sure you give viewing access to the link created). Upload the “.txt” file showing the link to your video. You must explain your work by speaking in English. First introduce yourself in few sentences in the beginning.

Deadline: 08/03/2026 23:59